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EXPERIMENT 3

AIM:

Apply Decision Tree and Random Forest for classification tasks

THEORY

1. Dataset Source

The dataset used for this experiment is the **Heart Disease Dataset** available on Kaggle.
Dataset Link:
https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset

The dataset is originally derived from the UCI Machine Learning Repository and is widely used for binary classification problems in medical diagnosis.

2. Dataset Description

The Heart Disease dataset contains medical attributes of patients and is used to predict the presence of heart disease.

Dataset Characteristics

- Total Records (Original): 1025
- Records after removing duplicates: 302
- Total Features: 13 input features + 1 target variable
- Target Type: Binary Classification (0 = No Disease, 1 = Disease)
- No missing values

Feature Description

Feature	Description
age	Age of patient
sex	Gender (1 = male, 0 = female)

cp	Chest pain type (0–3)

trestbps	Resting blood pressure
chol	Serum cholesterol
fbs	Fasting blood sugar (>120 mg/dl)
restecg	Resting electrocardiographic results
thalach	Maximum heart rate achieved
exang	Exercise induced angina
oldpeak	ST depression induced by exercise
slope	Slope of peak exercise ST segment
ca	Number of major vessels (0–4)
thal	Thalassemia type
target	0 = No heart disease, 1 = Heart disease

This dataset is suitable for supervised learning classification tasks.

3. Mathematical Formulation of the Algorithms

A) Decision Tree

Decision Tree is a supervised learning algorithm that splits the dataset based on feature values to maximize class purity.

The splitting criterion used is **Gini Impurity**:

$$Gini = 1 - \sum_{i=1}^{C} p_i^2$$

Where:

- p_i = probability of class i
- C = number of classes

The algorithm selects the feature that minimizes Gini impurity at each split.

B) Random Forest

Random Forest is an ensemble learning method that builds multiple Decision Trees and combines their predictions.

For classification:

$$Final\ Prediction = Majority\ Vote(T_1, T_2, ..., T_n)$$

Where:

- Ti represents individual decision trees
- n = number of trees

Each tree is trained on:

- A bootstrap sample of data
- A random subset of features

This reduces variance and overfitting.

4. Algorithm Limitations

Decision Tree Limitations

- Prone to overfitting
- Sensitive to small data variations
- High variance
- May create biased trees if classes are imbalanced

Random Forest Limitations

- More computationally expensive
- Less interpretable than Decision Tree
- Large number of trees increases memory usage
- Slower training compared to single tree

Random Forest may not perform well when:

- Dataset is extremely small
- Features are highly correlated
- Real-time prediction with strict latency constraints

5. Methodology / Workflow

Step 1: Data Collection

- Download dataset from Kaggle
- Load dataset into Google Colab

Step 2: Data Preprocessing

- Remove duplicate records
- Check missing values
- Separate features and target
- Apply train-test split (80:20)
- Apply feature scaling

Step 3: Model Training

- Train Decision Tree classifier
- Train Random Forest classifier

Step 4: Model Evaluation

- Calculate Accuracy
- Generate Confusion Matrix
- Generate Classification Report
- Plot ROC Curve

6. Performance Analysis

Decision Tree Results

- Accuracy: 73.77%
- Moderate performance
- Lower recall for disease class (0.69)

Random Forest Results

- Accuracy: 83.60%
- Better precision and recall
- Higher recall (0.90) for disease class

Interpretation

Random Forest outperformed Decision Tree due to:

- Ensemble learning
- Reduced overfitting
- Better generalization

In medical diagnosis, recall for the disease class is very important. Random Forest achieved higher recall, making it more reliable.

7. Hyperparameter Tuning

Hyperparameter tuning was performed to improve model performance.

Decision Tree Tuned Parameters

- max_depth
- min_samples_split
- min_samples_leaf

Example:

- max_depth = 5 reduced overfitting
- Controlled tree complexity

Random Forest Tuned Parameters

- n_estimators (number of trees)
- max_depth
- max_features

Example:

- n_estimators = 200 improved stability
- max_depth = 6 improved generalization

Impact of Tuning

Model	Before Tuning	After Tuning
Decision Tree	Lower accuracy	Improved stability
Random Forest	Good accuracy	Better generalization

Hyperparameter tuning reduced overfitting and improved predictive performance.

Code & Output

```
# 1 Import Libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as
plt import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (accuracy_score, classification_report,
                             confusion_matrix, roc_curve, auc)
from sklearn.preprocessing import StandardScaler

# 2 Upload Dataset
from google.colab import files
uploaded = files.upload()

df = pd.read_csv("heart.csv")
```

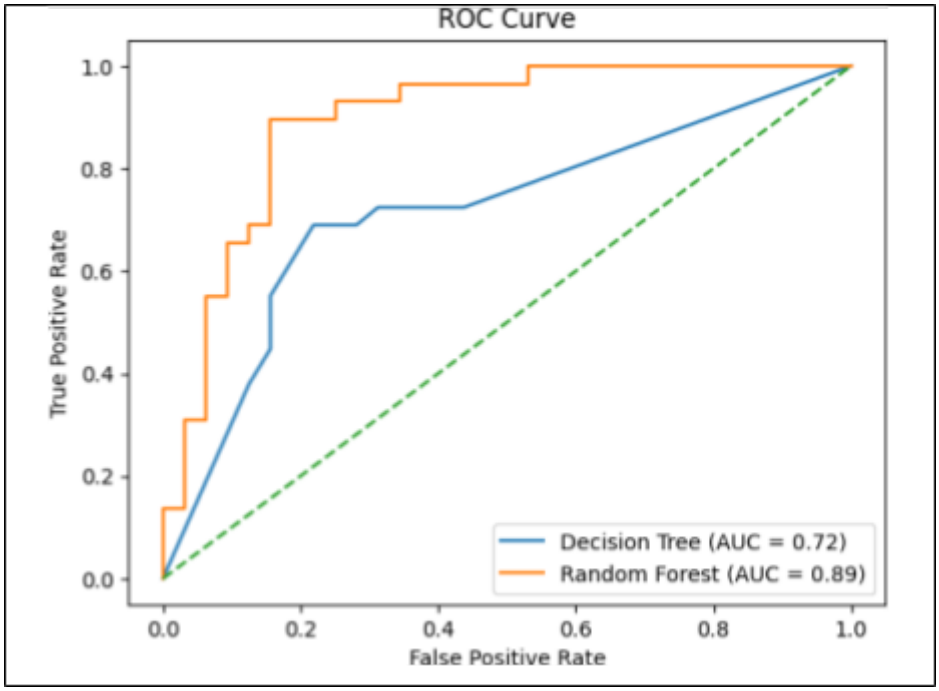
```
print("Original Shape:", df.shape)

# 3 Remove Duplicate Records (IMPORTANT FIX)
df = df.drop_duplicates()
print("Shape After Removing Duplicates:", df.shape)
# 4 Check Missing Values
print("\nMissing Values:\n", df.isnull().sum())

# 5 Define Features & Target
X = df.drop("target", axis=1)
y = df["target"]

# 6 Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 7 Feature Scaling
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

#
====================================
# DECISION TREE
# ====================================

dt = DecisionTreeClassifier(max_depth=5, random_state=42)
dt.fit(X_train, y_train)

y_pred_dt = dt.predict(X_test)

print("\n===== DECISION TREE RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_dt))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred_dt))
print("\nClassification Report:\n", classification_report(y_test, y_pred_dt))

#
====================================
# 🌲 RANDOM FOREST
# ====================================

rf = RandomForestClassifier(n_estimators=200, max_depth=6, random_state=42)
rf.fit(X_train, y_train)
```

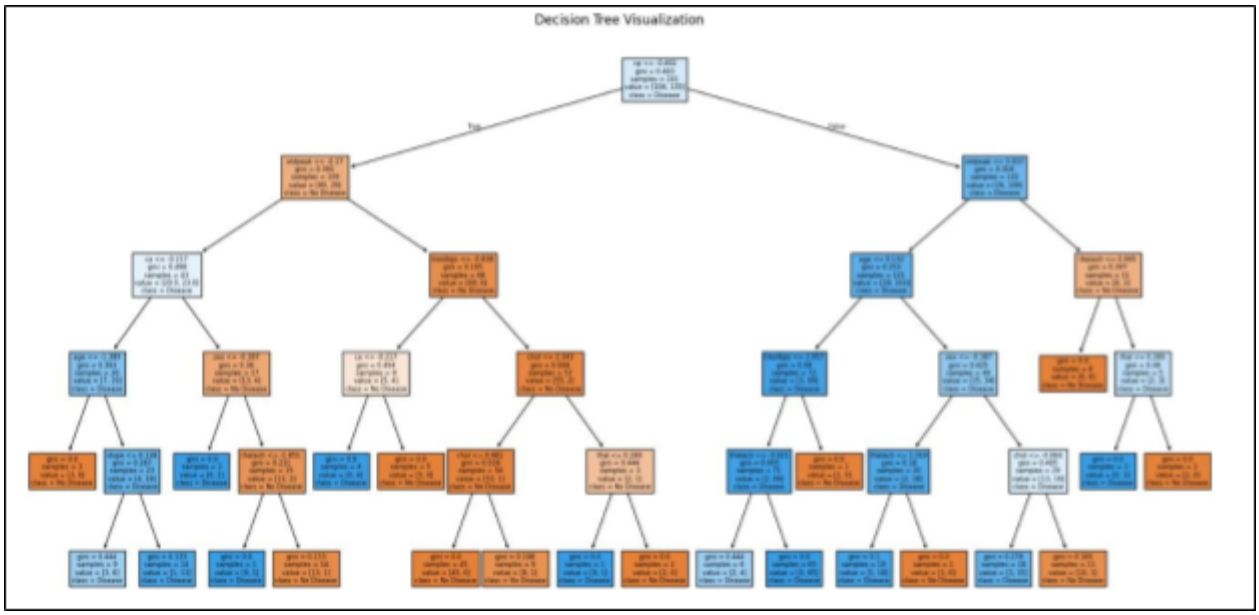
```
y_pred_rf = rf.predict(X_test)

print("\n===== RANDOM FOREST RESULTS =====")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred_rf))
print("\nClassification Report:\n", classification_report(y_test, y_pred_rf))

# ======================================
# Accuracy Comparison
# ======================================

dt_acc = accuracy_score(y_test, y_pred_dt)
rf_acc = accuracy_score(y_test, y_pred_rf)

print("\n===== MODEL COMPARISON =====")
print("Decision Tree Accuracy :", dt_acc)
print("Random Forest Accuracy :", rf_acc)

# ======================================
# ROC Curve
# ======================================

dt_probs = dt.predict_proba(X_test)[:,1]
rf_probs = rf.predict_proba(X_test)[:,1]

fpr_dt, tpr_dt, _ = roc_curve(y_test, dt_probs)
fpr_rf, tpr_rf, _ = roc_curve(y_test, rf_probs)

plt.figure()
plt.plot(fpr_dt, tpr_dt, label="Decision Tree (AUC = %0.2f)" % auc(fpr_dt, tpr_dt))
plt.plot(fpr_rf, tpr_rf, label="Random Forest (AUC = %0.2f)" % auc(fpr_rf, tpr_rf))
plt.plot([0,1],[0,1],'--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

# ======================================
# Decision Tree Visualization
# ======================================

plt.figure(figsize=(20,10))
plot_tree(dt, feature_names=df.columns[:-1],
        class_names=["No Disease","Disease"],
        filled=True)
```

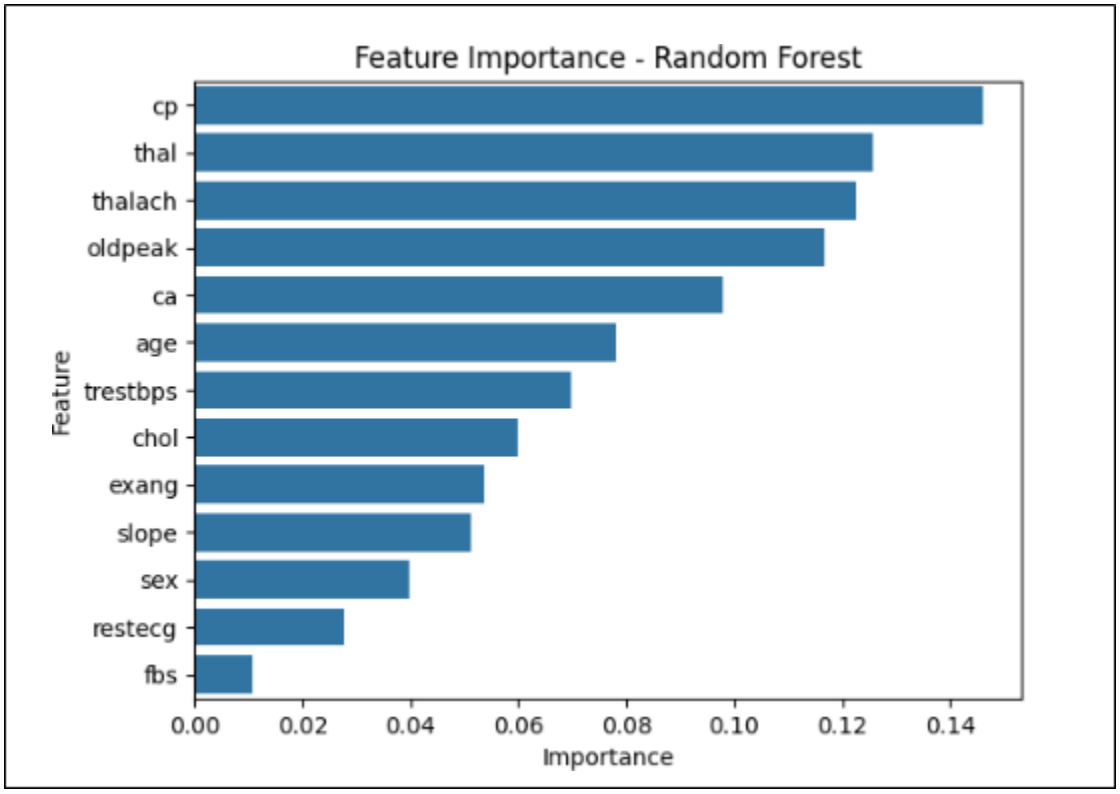
```
plt.title("Decision Tree Visualization")
plt.show()

# ======================================
# Feature Importance (Random Forest)
# ======================================

feature_importance = pd.DataFrame({
    'Feature': df.columns[:-1],
    'Importance':
    rf.feature_importances_
}).sort_values(by='Importance', ascending=False)

plt.figure()
sns.barplot(x='Importance', y='Feature', data=feature_importance)
plt.title("Feature Importance - Random Forest")
plt.show()
```

```
...  Choose Files  heart.csv
     heart.csv(text/csv) - 38114 bytes, last modified: 2/21/2026 - 100% done
     Saving heart.csv to heart (1).csv
     Original Shape: (1025, 14)
     Shape After Removing Duplicates: (302, 14)

     Missing Values:
      age         0
     sex         0
     cp          0
     trestbps    0
     chol        0
     fbs         0
     restecg     0
     thalach     0
     exang       0
     oldpeak     0
     slope       0
     ca          0
     thal        0
     target      0
     dtype: int64

     ===== DECISION TREE RESULTS =====
     Accuracy: 0.7377049180327869

     Confusion Matrix:
      [[25  7]
      [ 9 20]]
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.74      0.78      0.76        32
           1       0.74      0.69      0.71        29

    accuracy                           0.74        61
   macro avg       0.74      0.74      0.74        61
weighted avg       0.74      0.74      0.74        61


===== RANDOM FOREST RESULTS =====
Accuracy: 0.8360655737704918

Confusion Matrix:
 [[25  7]
 [ 3 26]]

Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.78      0.83        32
           1       0.79      0.90      0.84        29

    accuracy                           0.84        61
   macro avg       0.84      0.84      0.84        61
weighted avg       0.84      0.84      0.84        61


===== MODEL COMPARISON =====
Decision Tree Accuracy : 0.7377049180327869
Random Forest Accuracy : 0.8360655737704918
```







Conclusion

In this experiment, Decision Tree and Random Forest algorithms were applied to the Heart Disease dataset. After removing duplicate records, Random Forest achieved higher accuracy (83.60%) compared to Decision Tree (73.77%). Due to its ensemble nature and better recall for the disease class, Random Forest proved to be more suitable for heart disease prediction.